



Neural Networks and Training

Model Evaluation

Deploying a Model

Anomaly Detection

- ✓ **Video:** Anomaly Detection
8 min
- ✓ **Video:** Industrial Embedded Machine Learning Demo
4 min
- ✓ **Reading:** Anomaly Detection
5 min
- ✓ **Practice Quiz:** Anomaly Detection
1 question

Project and Review

- ✓ **Reading:** Project - Motion Detection
2h
- ▶ **Video:** Module Review
3 min
- 📄 **Quiz:** Motion Classification and Anomaly Detection
10 questions
- 💬 **Discussion Prompt:** Share Your Motion Detection Project!
15 min



Project - Motion Detection

Introduction

Welcome to the first project of the course! In this project, we will ask you to use the previous modules to create a fully-functioning embedded machine learning system performing feature extraction, training a model, and deploying that model to a device.

We will classify the motion and vibration data from a machine of your choice in an industrial environment. We want to be able to determine if a machine is operating normally or if it is showing signs of failure.

Please note that at this time, we do not have a way to grade your project for the course. However, we strongly encourage you to go through these steps to learn as much as possible about machine learning as well as using the Edge Impulse tool.

Required Hardware

You may use either your smartphone or [Arduino Nano 33 BLE Sense](#) to collect data with that device for data collection and deployment. Please do not mix the data from the smartphone to the Arduino).

We recommend using the Arduino board, if you have access to it. There are many ways to modify the Arduino code. We encourage you to try this challenge to get the most out of the data.

You will also need some tape (recommended: electrical tape) to secure the device to the machine.

Setup

Before we start collecting data, we must first figure out what we want to monitor. We will use a “magic wand” demo that classified person-made hand movements. In this project, we will use the “magic wand” to monitor different operating modes and look for anomalies. While the “magic wand” is a fun demo, it is not a realistic industrial uses for embedded machine learning in this project.

To begin, choose a machine in your home, office, or school that produces a specific sound and classify. Here are some ideas of things you may want to monitor:

- Washing machine
- Blender
- Air compressor
- Refrigerator
- Air conditioner
- Keyboard (typing)